

Day 93: Probability Fundamentals

1. Day number

Day 93

2. Topic name

Probability and Simple Events

Probability is a way to describe **how likely something is to happen**.

3. Connection

Yesterday, descriptive statistics helped you summarize **data that has already been observed**.

Today, probability helps you describe **uncertainty about outcomes that could happen**.

A simple way to remember the difference:

Statistics	→ What happened in the data?
Probability	→ What could happen, and how likely is it?

4. Important topics

Today you will learn:

- **Experiment**
- **Outcome**
- **Event**
- Probability between **0 and 1**
- **Complement** of an event

For simple cases where every outcome is equally likely:

Probability = number of favorable outcomes total number of possible outcomes
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5. Foundational notes

Probability always lies between:

0 and 1

Think of the scale like this:

0 ----- 0.5 ----- 1
Impossible uncertain Certain

For example:

Probability = 0

means the event cannot happen.

Probability = 1

means the event will definitely happen.

And:

Probability = 0.5

means a 50% chance.

Probability does **not** guarantee what will happen in one individual attempt. It describes likelihood.

6. Experiment, outcome, and event

Experiment

An **experiment** is an action whose result is uncertain.

Examples:

- flipping a coin
- rolling a die
- randomly selecting a customer
- randomly selecting a product

Suppose we roll a normal six-sided die.

That is the **experiment**.

Outcome

An **outcome** is one possible result of the experiment.

For a die:

1, 2, 3, 4, 5, 6

Each number is an individual outcome.

Event

An **event** is one or more outcomes that we are interested in.

For example:

| Rolling an even number.

The successful outcomes are:

2, 4, 6

So the event contains three possible outcomes.

7. Probability using simple examples

Coin example

A normal coin has two possible outcomes:

Heads
Tails

Suppose we want:

| Probability of getting Heads.

There is:

1 favorable outcome
2 total outcomes

Therefore:

$P(\text{Heads}) = 1 / 2$
 $= 0.5$

So the probability is:

0.5 or 50%

Dice example

Suppose we roll a fair six-sided die:

1, 2, 3, 4, 5, 6

What is the probability of rolling 4?

There is one favorable outcome:

4

out of six possible outcomes.

Therefore:

$$P(4) = 1 / 6$$

Approximately:

0.167

or:

16.7%

Business example

Imagine a box contains five customer orders:

Order 1 → Delivered
Order 2 → Pending
Order 3 → Delivered
Order 4 → Delivered
Order 5 → Pending

Suppose one order is selected randomly.

There are:

3 delivered orders
5 total orders

So:

$$\begin{aligned} P(\text{Delivered}) &= 3 / 5 \\ &= 0.6 \end{aligned}$$

That means:

60% probability

This is a simplified example where every order has an equal chance of being selected.

8. Easy example

Suppose a bag contains four cards:

Red
Blue
Red
Green

One card is selected randomly.

What is the probability of selecting a **red card**?

First count the total cards:

4

Count the favorable outcomes:

2 red cards

Therefore:

$$\begin{aligned} P(\text{Red}) &= 2 / 4 \\ &= 0.5 \end{aligned}$$

So:

Probability of red = 0.5 = 50%

9. Complement

The **complement** of an event means:

| The event does not happen.

Suppose:

$$P(\text{Rain}) = 0.3$$

Then:

$$P(\text{No Rain}) = 0.7$$

Why?

Because either it rains or it does not rain, so their probabilities must total **1**.

The basic complement rule is:

$$P(\text{not } A) = 1 - P(A)$$

For example:

$$P(\text{Rain}) = 0.3$$

$$\begin{aligned} P(\text{No Rain}) \\ &= 1 - 0.3 \\ &= 0.7 \end{aligned}$$

A useful check is:

$$P(A) + P(\text{not } A) = 1$$

10. Problem statement

A fair six-sided die contains:

1, 2, 3, 4, 5, 6

Calculate the probability of:

1. Rolling a
2. Rolling an even number
3. Rolling a number greater than
4. Not rolling a

Do the calculations manually before using Python.

11. Concepts used

This exercise uses:

- experiment
- sample of possible outcomes
- outcome
- event
- favorable outcomes
- total outcomes
- simple probability
- complement
- fractions
- decimals
- percentages

12. Thought process

When solving a beginner probability problem, follow this pattern:

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What is the experiment?
↓
What outcomes are possible?
↓
Which outcomes satisfy my event?
↓
Count favorable outcomes
↓
Count total outcomes
↓
favorable / total
↓
Convert to decimal or percentage if needed

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For example, suppose the event is:

| Roll an even number.

Start with all outcomes:

1, 2, 3, 4, 5, 6

Then identify only the outcomes belonging to the event:

2, 4, 6

Now compare the number of favorable outcomes with the total number of outcomes.

13. Beginner-friendly formulas and calculations

For equally likely outcomes:

$P(\text{Event}) =$
 Favorable outcomes / Total possible outcomes

Suppose there are 2 favorable outcomes from 8 possible outcomes:

$P(\text{Event}) = 2 / 8$

Simplify:

$= 1 / 4$

As a decimal:

$= 0.25$

As a percentage:

$$= 25\%$$

So all of these describe the same probability:

$$1/4 = 0.25 = 25\%$$

For a complement:

$$P(\text{not } A) = 1 - P(A)$$

If:

$$P(A) = 0.25$$

then:

$$\begin{aligned} P(\text{not } A) \\ &= 1 - 0.25 \\ &= 0.75 \end{aligned}$$

14. Percentage vs probability

Probability is commonly written as either a decimal or a percentage.

For example:

$$\text{Probability} = 0.75$$

means the same thing as:

$$75\%$$

To convert a probability to a percentage:

$$\text{probability} \times 100$$

Example:

$$0.4 \times 100 = 40\%$$

To convert a percentage back to probability:

percentage / 100

Example:

$$40 / 100 = 0.4$$

A useful reference is:

Probability	Percentage	Meaning
0	0%	Impossible
0.25	25%	One-quarter chance
0.5	50%	Half chance
0.75	75%	Three-quarter chance
1	100%	Certain

15. Easy edge cases

Impossible event

Suppose we roll a normal six-sided die.

What is the probability of rolling 10?

The die contains:

1, 2, 3, 4, 5, 6

There are no favorable outcomes.

Therefore:

$$P(10) = 0 / 6 = 0$$

So an impossible event has:

Probability = 0

Certain event

What is the probability of rolling a number between 1 and 6?

Every possible outcome satisfies this condition.

6 favorable outcomes
6 total outcomes

Therefore:

$$6 / 6 = 1$$

So a certain event has:

Probability = 1

16. Expected calculations

For a simple problem, your working should generally look like this:

Possible outcomes:

{...}

Favorable outcomes:

{...}

Number of favorable outcomes = ?

Total number of outcomes = ?

Probability =
favorable / total

Decimal = ?

Percentage = ?%

For complements:

$P(\text{event}) = \dots$

$P(\text{not event})$
 $= 1 - P(\text{event})$

The most important rule for today is:

$$0 \leq \text{Probability} \leq 1$$

17. Hint only

For the die exercise:

1, 2, 3, 4, 5, 6

For **rolling a 3**, count how many 3s appear among the six possibilities.

For **rolling an even number**, first identify:

2, 4, 6

For **greater than 4**, identify which outcomes satisfy:

number > 4

For **not rolling a 6**, you can either count all outcomes except **6**, or use the complement:

$$P(\text{not } 6) = 1 - P(6)$$

Keep everything in the form:

favorable outcomes

total outcomes

before converting your answer to a decimal or percentage.